Day 27 | Wednesday

27

December 16, 2026

Readiness check: 12 core questions

9:00 AM–10:15 AM • Math Review • 75 min

Corrections and a personal practice plan

10:30 AM–12:00 PM • Math Review • 90 min

Week 8 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Readiness check: 12 core questions

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:15 AM	**Teach**	5
9:15 AM–9:20 AM	**Directions**	5
9:20 AM–10:00 AM	**Independent check**	40
10:00 AM–10:10 AM	**Collect & reflect**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math Review • Math Week 8, Lesson 3

10:30 AM–12:00 PM • 90 min

Corrections and a personal practice plan

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Math Review • Math Week 8, Lesson 4

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Readiness check: 12 core questions

Today’s goal: Demonstrate independent calculation, units, working and checking across the core course skills.

Identify one target skill and the checking habit to use today.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Set in order: 4A Q14(a) p.36; 5A Q12 first value p.46; 6B Q2 p.57; 6A Q19 p.56; 7A Q11 p.68; 8C Q1 p.81; 9A Q1 p.85; 10B Q11 p.96; 11A Q7 p.103; 12A Q10 p.115; 13A Q6,14 p.128.

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Recall and reconnect

Confirm access needs, calculator model and permitted formula/conversion sheets. Explain this is not an official KPU entry exam or pass standard.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Set in order: 4A Q14(a) p.36; 5A Q12 first value p.46; 6B Q2 p.57; 6A Q19 p.56; 7A Q11 p.68; 8C Q1 p.81; 9A Q1 p.85; 10B Q11 p.96; 11A Q7 p.103; 12A Q10 p.115; 13A Q6,14 p.128.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Set up the independent response record

1. Prepare a numbered 1–12 response page in the exact order of the listed book questions.
2. For each response, retain a diagram where useful, the formula, units, calculation and final check.
3. Use the scheduled five-minute response example to see this format; it is a different example, not a scored item.
4. During Directions, clarify wording and rounding only. Do not solve readiness items together.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Keep the readiness attempt independent

1. Use the chosen calculator, Textbook formula sheet p.361 and conversion factors pp.362–363 under the stated conditions.

2. Keep Numeracy Hub worked answers and Shape Canvas answer displays closed; do not use tool hints or partner solutions.

3. Attempt all taught core items in the scheduled independent period. Any assistance is recorded separately; deferred skills are marked not yet taught.

4. Submit your working and identify two uncertain items during Collect & reflect. This is formative feedback, not an official KPU entry exam or institutional pass standard.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Unscored example: show a complete response

P = 2 × (3 + 2)
P = 10 m

Check: 3 + 2 + 3 + 2 = 10 m

Record a labelled sketch, formula, substitution, answer, units and check.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Directions for the readiness check

Clarify directions and rounding without solving any scored item. Learners set up a numbered 1–12 response page.

Set up a numbered response page. Clarify directions only; do not work with a partner on test items.

Resources: Set in order: 4A Q14(a) p.36; 5A Q12 first value p.46; 6B Q2 p.57; 6A Q19 p.56; 7A Q11 p.68; 8C Q1 p.81; 9A Q1 p.85; 10B Q11 p.96; 11A Q7 p.103; 12A Q10 p.115; 13A Q6,14 p.128.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Independent readiness check

Complete the twelve listed questions independently in 40 minutes. Try to attempt all core items; record help separately. If a skill was deferred, mark it not yet taught rather than assigning a false readiness judgement.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Set in order: 4A Q14(a) p.36; 5A Q12 first value p.46; 6B Q2 p.57; 6A Q19 p.56; 7A Q11 p.68; 8C Q1 p.81; 9A Q1 p.85; 10B Q11 p.96; 11A Q7 p.103; 12A Q10 p.115; 13A Q6,14 p.128.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 1 of 12 | 4A Q14a

Add. (a) $\frac{5}{16} + \frac{3}{4}$

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 2 of 12 | 5A Q12 — first value

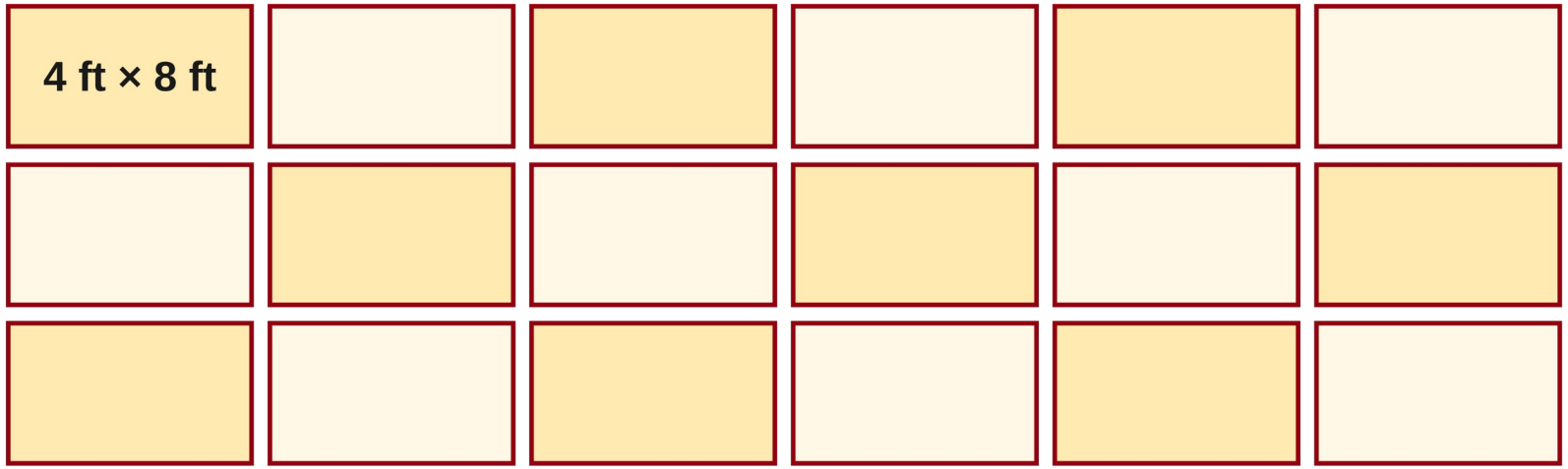
Change $\frac{7}{8}$ to a decimal.

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 3 of 12 | 6B Q2

A fabrication job needs 18 sheets of 4 ft × 8 ft plate. Add a 15% allowance and round up to whole sheets.



18 sheets

Add 15%

Whole sheets: ?

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 4 of 12 | 6A Q19

At a scale of 1 : 40, a line measures 85 mm. How long is the real length?

On the drawing



Scale 1 : 40

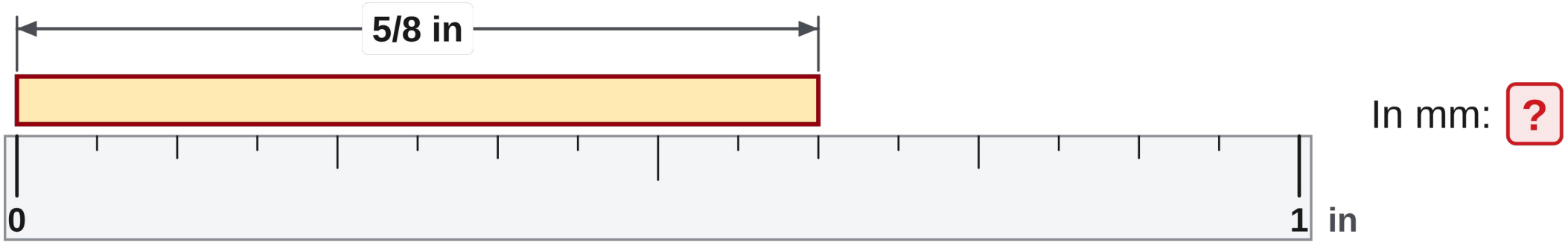
Real length: ?

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 5 of 12 | 7A Q11

Convert 5/8 in to mm.

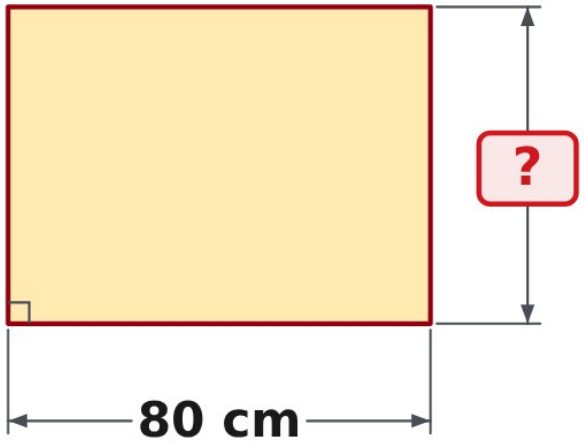


Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 6 of 12 | 8C Q1

The rectangle has an area of 4,800 cm². Find the width.

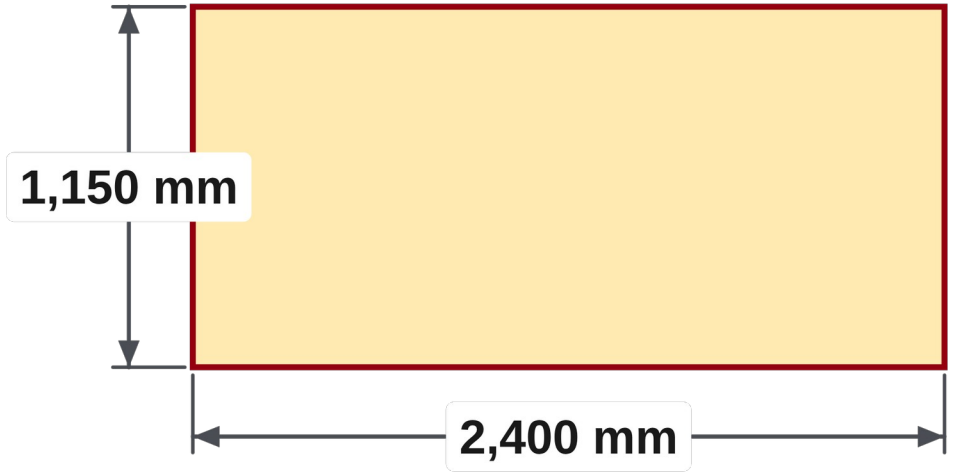


Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 7 of 12 | 9A Q1

Find the perimeter of a rectangle 2,400 mm by 1,150 mm.



Perimeter: ?

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 8 of 12 | 10B Q11

A 300 mm by 200 mm access hole is cut in a 1,200 mm by 800 mm cover plate. Find the area of plate left in mm² and in m².

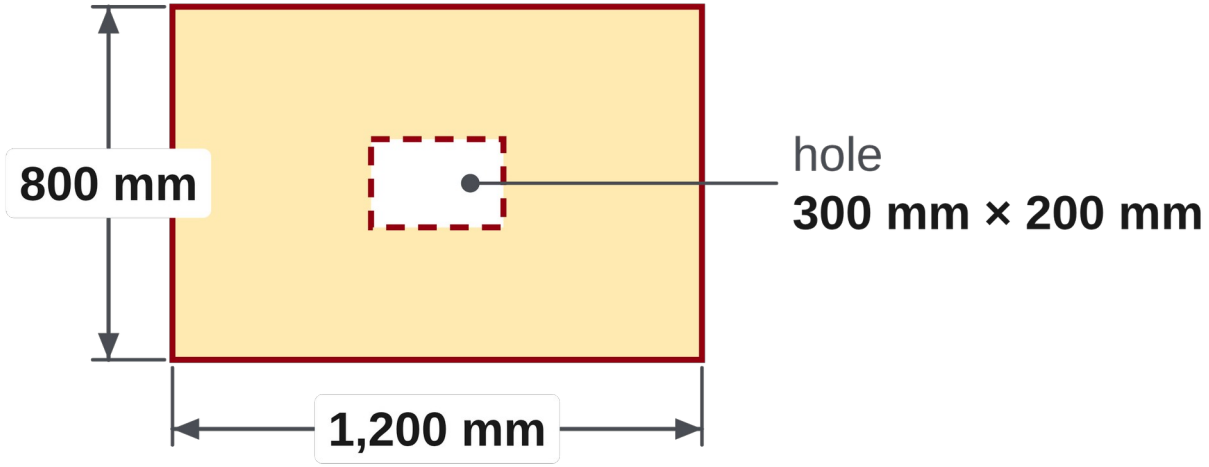


Plate left in mm²: ?

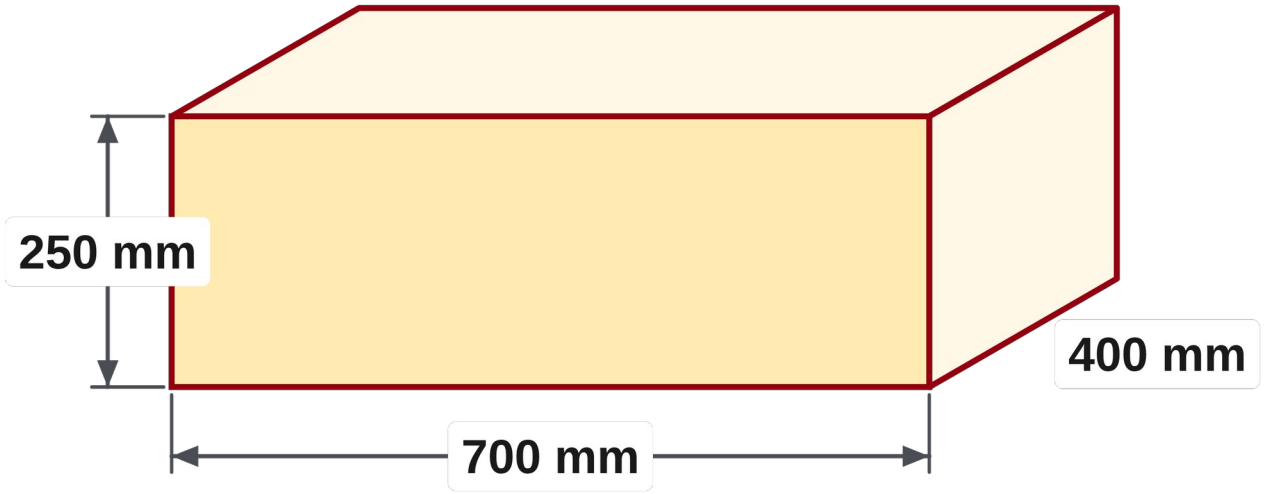
Plate left in m²: ?

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 9 of 12 | 11A Q7

Box 700 mm by 400 mm by 250 mm. Find the volume in m³.



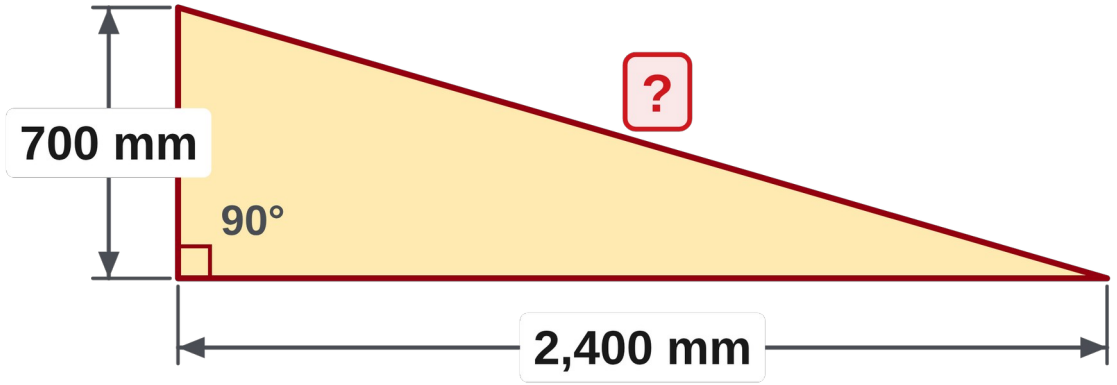
Volume in m³: ?

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 10 of 12 | 12A Q10

Legs of 700 mm and 2,400 mm. Find the hypotenuse.

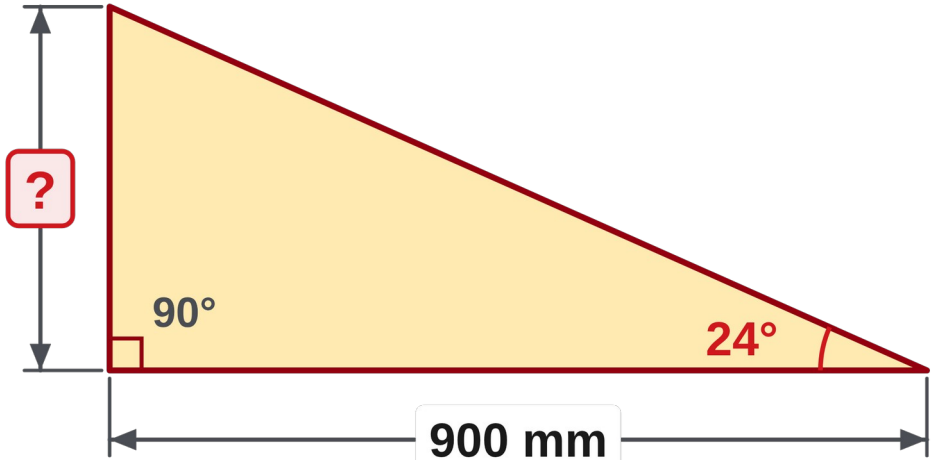


Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 11 of 12 | 13A Q6

Adjacent 900 mm, angle 24°. Find the opposite side to two decimal places.



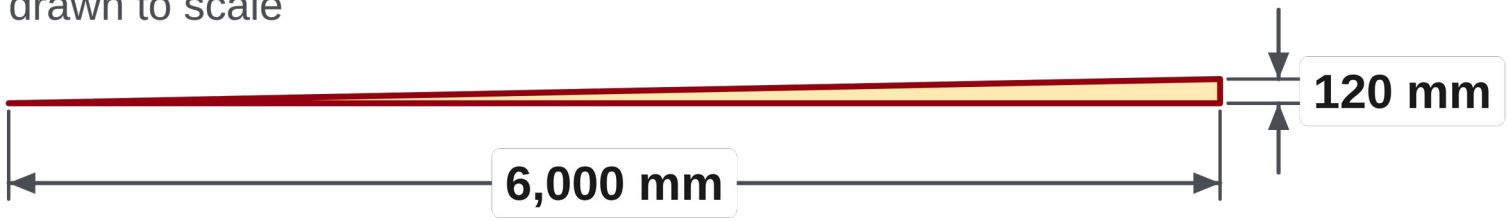
Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Readiness question 12 of 12 | 13A Q14

A slope rises 120 mm over a run of 6,000 mm. Find the percent grade and the angle to two decimal places.

drawn to scale



Percent grade: ?

Angle: ?

Independent readiness check. Record all working on your numbered response page.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Collect and reflect

Submit your work and identify two items you were uncertain about. Describe whether the difficulty involved the concept, operation, calculator entry or units.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Set in order: 4A Q14(a) p.36; 5A Q12 first value p.46; 6B Q2 p.57; 6A Q19 p.56; 7A Q11 p.68; 8C Q1 p.81; 9A Q1 p.85; 10B Q11 p.96; 11A Q7 p.103; 12A Q10 p.115; 13A Q6,14 p.128.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

Resources: Set in order: 4A Q14(a) p.36; 5A Q12 first value p.46; 6B Q2 p.57; 6A Q19 p.56; 7A Q11 p.68; 8C Q1 p.81; 9A Q1 p.85; 10B Q11 p.96; 11A Q7 p.103; 12A Q10 p.115; 13A Q6,14 p.128.

Textbook formula sheet p.361; conversion factors pp.362–363 permitted for this proposed formative check

Break

10:15 AM–10:30 AM

Return ready for: Corrections and a personal practice plan

This break is excluded from teaching minutes.

Corrections and a personal practice plan

Today’s goal: Correct a demonstrated error, solve a fresh problem of the same type and plan continued practice.

Identify one target skill and the checking habit to use today.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Choose missed types from the readiness set. Fresh checks: 4A Q15(b) p.36; 7A Q12 p.68; 10A Q6 p.93; 12A Q12 p.115; 13A Q10 p.128.

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Recall and reconnect

Return the readiness work privately; ask each learner to choose two errors to understand rather than only counting a score.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Choose missed types from the readiness set. Fresh checks: 4A Q15(b) p.36; 7A Q12 p.68; 10A Q6 p.93; 12A Q12 p.115; 13A Q10 p.128.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Explain the step that failed

1. Use your private readiness feedback to choose two errors to understand, rather than only counting a score.

2. Keep the correction record in order: original method → why it failed → corrected working → independent retry.

3. Use the relevant Textbook chapter, formula sheet p.361 and glossary pp.364–368 to make the explanation precise.

4. Include a diagram and units where relevant. Pair-check one correction, then use the scheduled fresh checks to show the method independently.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Match the fresh attempt to the repaired skill

1. Choose from the listed fresh checks according to the skills shown by your readiness errors; they are not a new requirement to complete every topic.

2. A fresh problem of the same type tests the method without relying on a remembered answer.

3. If the five-question Hub set is used, select only taught weak topics and keep worked answers off until after the attempt.

4. Your practice plan names two skills, exact book pages, tool modules and a concrete next practice step.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Explain why the first method failed

Possible demonstration A • Regroup before subtraction.
5 1/4 − 2 3/8 = 4 10/8 − 2 3/8 = 2 7/8.
Check: 2 7/8 + 2 3/8 = 5 1/4. Explain why the whole was regrouped.

Possible demonstration B • Convert the entire mixed number.
1 3/4 in = 1.75 in; 1.75 × 25.4 = 44.45 mm.
Check: 44.45 ÷ 25.4 = 1.75 in—not 1.34 in.

Instructor • Choose the two most common actual errors; use different numbers.
Together • Pair-check one correction while the instructor supports the highest-need group. Require a diagram and units where relevant.

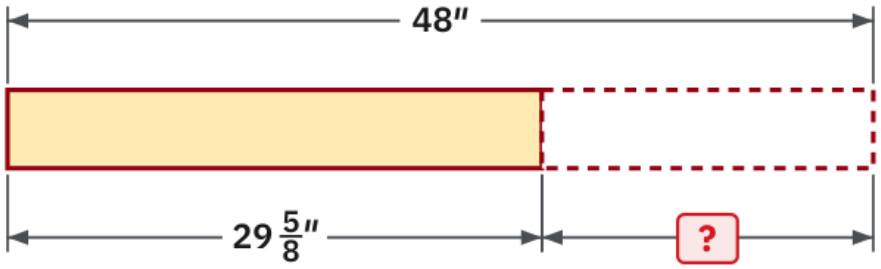
These are fresh classroom demonstrations, not the independent retry questions.

Relevant Textbook chapters selected from readiness errors • Explain the change, do not copy an answer

Textbook | page 99

Example 4.10 What is left of the bar?

A piece $29\frac{5}{8}$ in long is cut from a 48 in bar. Ignore the saw cut. How long is the offcut?



SOLUTION

1. There is no fraction in 48 to take $\frac{5}{8}$ from, so borrow 1: $48 = 47\ \frac{8}{8}$.
2. Fractions: $\frac{8}{8} - \frac{5}{8} = \frac{3}{8}$.
3. Whole numbers: $47 - 29 = 18$.

Use for a fraction priority: regrouping and an offcut diagram.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Textbook | page 99

CHECK BY ADDING

Add your answer to the number you took away. You should get the number you started with:

$18\frac{3}{8} + 29\frac{5}{8} = 48.$

Use for a fraction priority: verify subtraction by adding back.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Spot the mistake. Explain the correction.

A correction replaces the wrong result with the answer-key result and says “I understand.” What evidence is missing?

Discuss the method before checking the answer.

Explain why the original method failed, retain corrected working, and complete a fresh question in the priority skill independently with a checking strategy. The new attempt provides evidence that the method has improved.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Work together

Learners pair-check one correction while the instructor supports the highest-need group. Require a diagram and units where relevant.

Explain the method and check directions; work with a partner only where permitted.

Resources: Choose missed types from the readiness set. Fresh checks: 4A Q15(b) p.36; 7A Q12 p.68; 10A Q6 p.93; 12A Q12 p.115; 13A Q10 p.128.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Your scheduled practice

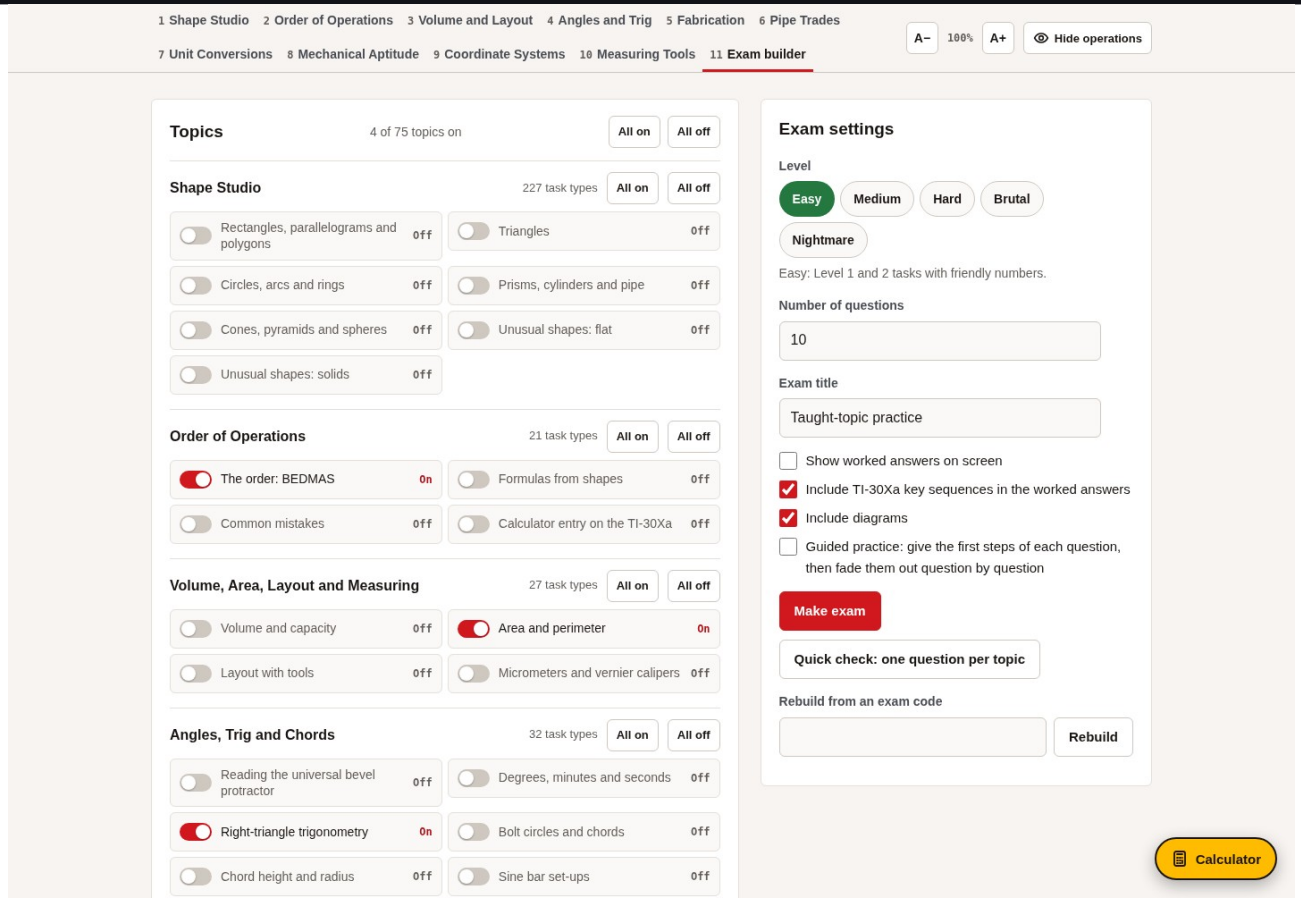
Correct missed items, then solve fresh checks matching the same skills. Create a five-question Numeracy Hub set if appropriate; keep its code. Write a personal plan naming two skills, book pages and tool modules for continued study.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Choose missed types from the readiness set. Fresh checks: 4A Q15(b) p.36; 7A Q12 p.68; 10A Q6 p.93; 12A Q12 p.115; 13A Q10 p.128.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Practise in Numeracy Hub and Shape Canvas

1. After the written attempt, use Shape Canvas to rebuild one missed diagram and connect its labels to the corrected method and units.

2. If appropriate during the scheduled practice, create a Numeracy Hub Exam builder set with only taught weak topics: Easy, 5 questions, worked answers off. Retain its code and written attempts.

3. Reveal worked answers only after the attempt to support corrections. Keep the code, corrected work and the two-skill plan with exact book pages and tool modules.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Practise from the Exercise Book | 4A Q15b

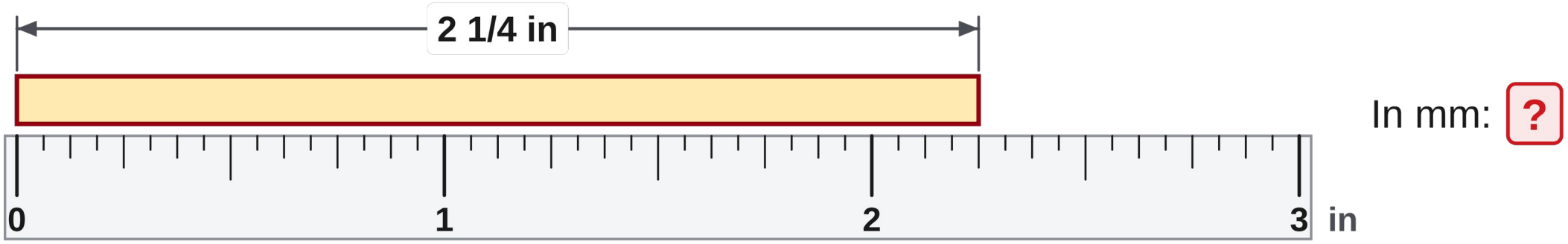
Subtract. (b) $9\frac{1}{8} - 4\frac{3}{4}$

Choose only fresh questions matching your two priority skills.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Practise from the Exercise Book | 7A Q12

Convert 2 1/4 in to mm.



Choose only fresh questions matching your two priority skills.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Check independently

Independently complete one fresh question in each priority skill and explain the checking strategy.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Choose missed types from the readiness set. Fresh checks: 4A Q15(b) p.36; 7A Q12 p.68; 10A Q6 p.93; 12A Q12 p.115; 13A Q10 p.128.

Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

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Textbook relevant chapters from readiness errors; formula sheet p.361; glossary pp.364–368